Calculus I Lesson 04

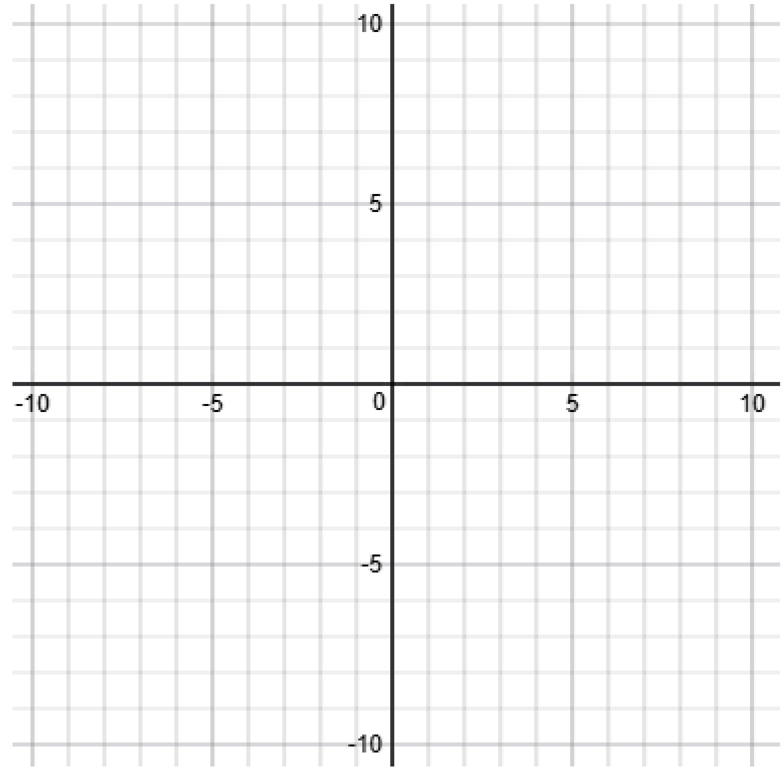
1) Find $\lim_{x \to 0} \frac{4x}{x^2 - x}$.

a) Factor $x^2 - x = x \cdot (\underline{\quad ? \quad})$

b) Simplify $\frac{4x}{x^2 - x} = \underline{\quad ? \quad}$

c) $\lim_{x \to 0} \frac{4x}{x^2 - x} = \underline{\quad ? \quad}$

d) Graph $f(x) = \frac{4x}{x^2 - x}$



2) Find $\lim_{x \to -3} \frac{x^2 - 4x + 3}{x^2 - 9}$.

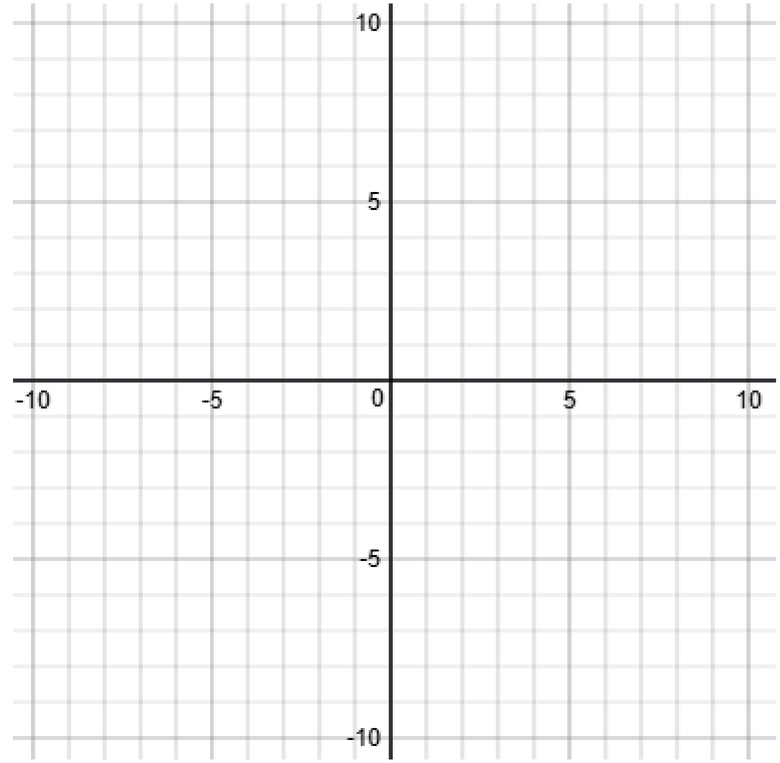
a) Factor $x^2 - 4x + 3 = \left(\underline{\quad ? \quad}\right) \cdot \left(\underline{\quad ? \quad}\right)$

b) Factor $x^2 - 9 = \left(\underline{\quad ? \quad}\right) \cdot \left(\underline{\quad ? \quad}\right)$

c) Simplify $\frac{x^2 - 4x + 3}{x^2 - 9} = \underline{\quad ? \quad}$

d) $\lim_{x \to -3} \frac{x^2 - 4x + 3}{x^2 - 9} = \underline{\quad ? \quad}$

e) Graph $f(x) = \frac{x^2 - 4x + 3}{x^2 - 9}$



3) Find $\lim_{x \to 5} \dfrac{x^2 - 4x - 5}{x - 5}$.

a) Factor $x^2 - 4x - 5 =$ $(\underline{\quad ? \quad}) \cdot (\underline{\quad ? \quad})$

b) Simplify $\dfrac{x^2 - 4x - 5}{x - 5} = \underline{\quad ? \quad}$

c) $\lim_{x \to 5} \dfrac{x^2 - 4x - 5}{x - 5} = \underline{\quad ? \quad}$

4) Find $\lim_{x \to 4} \dfrac{2x^2 - 5x - 12}{x - 4}$.

a) Factor $2x^2 - 5x - 12 = (\underline{\quad ? \quad}) \cdot (\underline{\quad ? \quad})$

b) Simplify $\dfrac{2x^2 - 5x - 12}{x - 4} = \underline{\quad ? \quad}$

c) $\lim_{x \to 4} \dfrac{2x^2 - 5x - 12}{x - 4} = \underline{\quad ? \quad}$

5) Find $\lim\limits_{x \to -5} \dfrac{4x^2+17x-15}{x+5}$.

a) Factor $4x^2+17x-15 = (\underline{\quad ? \quad}) \cdot (\underline{\quad ? \quad})$

b) Simplify $\dfrac{4x^2+17x-15}{x+5} = \underline{\quad ? \quad}$

c) $\lim\limits_{x \to -5} \dfrac{4x^2+17x-15}{x+5} = \underline{\quad ? \quad}$